Sun Wei Hao

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github.com/weihaowh weihaowh.github.io/Portfolio

Education

Singapore Institute of Technology **September 2022 – April 2026**
Bachelor of Science with Honours in Computer Science in Real-Time Interactive Simulation *Singapore*

Nanyang Polytechnic **April 2017 – March 2020**
Diploma in Mechatronics Engineering *Singapore*

Technical Skills

Languages: C++, C#, Python, C, TypeScript, JavaScript, GDScript, Kotlin, HTML/CSS
Engines & XR: Unreal Engine, Unity, Godot, OpenXR, XR/VR Development, Blender
Simulation: Real-Time Simulation & Modelling, Physics Simulation, Procedural Generation, Pathfinding & Navigation (NavMesh, A*), Geodetic Coordinate Systems, Digital Twins, Performance Optimisation
AI & Agents: Agentic Systems & Workflows, Model Context Protocol (MCP) & Agent Skills, LLM Integration, RAG Pipelines, Computer Vision (OpenCV), Machine Learning, Claude & AI Coding Tools
Engineering: Data Structures & Algorithms, OOP & Design Patterns, REST APIs, Multithreading, Automated Testing, CI/CD, Agile/Scrum
Version Control & DevOps: Git (GitHub, GitLab), Perforce Helix Core, Anchorpoint (3D version control), Docker, AWS, Firebase, Linux, VS Code

Experience

HTX (Home Team Science & Technology Agency) **July 2026 – Present**
Simulation Engineer *Singapore*

- Design and develop real-time simulations for Home Team training and research.
- Build interactive training scenarios that let officers practise operational procedures in a safe, repeatable virtual environment.

AETHER **May 2025 – April 2026**
Software Engineer *Singapore*

- Developed multiple real-time simulation systems using Unreal Engine, Unity, and Godot for training and research.
- Built a rule-based procedural simulation framework in Unreal Engine 5 using PCG, enabling scalable and deterministic environment generation for autonomous drone testing.
- Engineered state-machine-driven systems for aircraft and drone behaviours, supporting multi-phase operations and real-time interaction workflows.
- Developed a Precision Approach Radar (PAR) simulation, including aircraft state machines, radar visualisation, and landing guidance logic.
- Worked with geodetic coordinates, waypoint-based navigation, and real-time data visualisation.
- Optimised performance across simulations by managing AI density, rendering load, and update frequency to maintain stable FPS.

Sanmina **September 2019 – February 2020**
Robotics Engineer *Singapore*

- Received the FESTO Outstanding Project Award for a robotics and automation project.
- Reprogrammed Autonomous Intelligent Vehicle (AIV) routes for improved efficiency, mapped paths using LiDAR, and resolved localisation and mapping issues.
- Improved a production line's manufacturing process by 60%, benefiting 60+ production line employees.
- Developed computer vision-based quality inspection for conveyor belt systems using C++, enabling automated defect detection in the manufacturing line.
- Specialised in collaborative robots (cobots), with hands-on experience in programming, integration, and safety compliance for industrial automation.
- Wrote, reviewed, and edited technical documentation in accordance with template requirements.

Projects

EXODUS2 | *Game Development, Steam Release*

- First-person shooter released on Steam, where I contributed to post-processing effects, shaders, and gameplay scripting.
- Implemented and tuned post-processing effects to enhance visual atmosphere, including colour grading and screen effects.
- Worked on shader-related features to support visual feedback and gameplay presentation.
- Developed gameplay scripting logic

Seam-Carving | *C++, OpenCV*

- Implemented a content-aware image resizing technique using both dynamic programming and greedy algorithms.
- Preserved key image features by identifying and removing low-energy seams, supporting both vertical and horizontal seam removal.
- Minimised distortion in important regions while still achieving target dimensions efficiently.

Procedural Simulation Framework | *Unreal Engine 5, PCG, World Partition*

- Rule-driven procedural environment framework built in Unreal Engine 5 for large-scale simulation and autonomous drone testing
- Built a rule-driven procedural environment generation framework using UE5 PCG graphs for large-scale simulation.
- Implemented spline-based road generation and terrain-aware asset placement with collision checks
- Optimised performance for dense, high-realism scenes using instancing/HISM, LOD tuning, and streaming via World Partition.

PRISM (Police Response & Incident Simulation Module) | *Unity, C#, NavMesh*

- Urban traffic and behaviour simulation developed in Unity as part of the HackX Hackathon for police training and incident response scenarios.
- Implemented an urban traffic simulation, including traffic-light state machines and intersection control logic.
- Developed AI traffic agents with modular vehicle behaviours (stop/yield/lane-following) and obstacle-aware movement.
- Built pedestrian navigation using NavMesh, integrating crossing behaviour and signal-based synchronisation with traffic flow.

Precision Approach Radar (PAR) Simulation | *Godot, GDScript, Geodetic Waypoints*

- Radar-based air-traffic-control training simulation built in Godot to model precision approach guidance and landing workflows.
- Developed a radar-based approach-control training simulation with multi-aircraft traffic management and landing-request workflows.
- Implemented aircraft state machines, waypoint navigation, and geodetic flight paths loaded from JSON and converted to world coordinates.
- Built a PAR guidance interface with real-time deviation visualisation for localizer and glide slope alignment, plus scoring/handover logic.

Magique | *C++, C#*

- *Magique* is a 2D thriller side-scrolling platformer in which you play as Amare, a young boy trying to escape from an exploitative circus run by its ringleader, Malachi.
- Served as the main programmer in a team of 8, focusing on implementing core gameplay mechanics (movement, puzzles, combat) and interactive elements that shape the overall player experience.
- Collaborated closely with artists and designers to integrate animations, level designs, and narrative elements, ensuring consistent art style and immersive storytelling.

AstroCow | *C, C++*

- *AstroCow* is a game where you pilot a UFO to abduct farm animals and transport them safely back to your home planet. Avoiding asteroids and environmental hazards is critical for survival.
- Contributed as a programmer in a team of 5, implementing camera, spawning logic for asteroids and game logic, and building the overall win/lose flow and credits sequence.
- Utilized object pooling and scene management techniques for optimized performance on various hardware configurations.

Personal Expense Tracker App | *Kotlin, Jetpack Compose, OCR*

- Developed a lightweight budgeting tool to record expenses and display categorised spending charts.
- Integrated OCR capabilities for automatic receipt scanning, greatly simplifying data entry for users.